

PAPER_1209KK_Tier_KK_Solar_System

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UQFF Tier KK: Solar System Closures from 11 Frozen Primitives

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Date: May 2026

Result Ten heliocentric and terrestrial scales reproduced by $\pm$ -integer combinations of the eleven locked primitives, with no fitted constants and a sub-half-percent residual.

```
\begin{center}
\begin{tabular}{lllll}
\toprule
ID & quantity & predicted & observed & err \midrule
S684 & 1 AU  $\times 10^{10}$  m & 15.016 & 14.96 & 0.374% S685 &  $M_{\odot} \times 10^{29}$  kg & 19.838 & 19.89 & 0.261% S686 & Earth orbital  $v$  (km/s) & 29.7184 & 29.78 & 0.207% S687 &  $R_{\odot} \times 10^8$  m & 6.99417 & 6.96 & 0.491% S688 &  $M_J \times 10^{27}$  kg & 1.90458 & 1.898 & 0.347% S689 &  $R_{\oplus} \times 10^6$  m & 6.36349 & 6.371 & 0.118% S690 & Moon sidereal period (day) & 27.3213 & 27.32 & 0.005% S691 & sidereal year  $\times 100$  day & 3.64293 & 3.65256 & 0.264% S692 & Mars semi-major axis (AU) & 1.52051 & 1.524 & 0.229% S693 & Mercury year  $\times 10$  day & 8.82205 & 8.797 & 0.285%
\bottomrule
\end{tabular}
\end{center}
```

Cumulative Audit After Tier KK the master registry stands at **182/437**** audit-verifiable numeric closures (``master_closures.csv``),**

binned by accuracy in ``sigma_table.csv``:

```
\begin{center}
\begin{tabular}{lr}
\toprule
band & count \midrule
EXACT & 47
$<0.01\%$ & 18
$0.01\%$--$0.1\%$ & 60
$0.1\%$--$1\%$ & 54
$\geq 1\%$ & 3
\bottomrule
\end{tabular}
\end{center}
```

All 182 closures bridge to the 9-sector L_{UQFF} action via `bridge_map.csv`; the Tier-KK entries route through $\mathcal{L}_{\text{buoy}}$ (β_i), $\mathcal{L}_{\phi}$ (K_{Mex}), $\mathcal{L}_{\text{aether}}$ (F_{trz}), and the master channel-count constant $N_{\text{ch}}=9$.

Reproducibility

```
\begin{verbatim}
.\venv\Scripts\python.exe
_uqff_program.py --search --tier KK \ --targets _targets_KK.csv
--max-terms 4 --tol 0.005 .\venv\Scripts\python.exe _build_tier_KK.py
```

```
.\.venv\Scripts\python.exe _uqff_program.py --audit
```

```
.\.venv\Scripts\python.exe _uqff_program.py --sigma \end{verbatim}
```

Cumulative: **399 closures, 94 lockings** toward the
1000-closure / 100-locking target. Coverage 39.9%.